Name: Mohammad Abaid

Roll no: B-08

Branch:- AI&DS(B)

Experiment:- 04

AIM:-Data Wrangling with Pandas: Manipulate and transform data to suit the analysis need.

Q1. Explain the role of NumPy in the Python scientific computing ecosystem.

ANS:- NumPy (Numerical Python) is the core library for scientific computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a wide range of mathematical functions to operate on them. Its key roles include:

- Acting as the foundation for many other scientific libraries.
- Enabling efficient storage and manipulation of numerical data
- Supporting vectorized operations, reducing the need for loops.
- Serving as the base for libraries like Pandas, Matplotlib, and Scikit-learn.

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print("NumPy Array:", arr)
print("Array Type:", type(arr))
```

```
NumPy Array: [1 2 3 4 5]
Array Type: <class 'numpy.ndarray'>
```

Q2. Why is NumPy considered faster than pure Python computations?

ANS:- NumPy is faster than pure Python because:

- Vectorization: Operations are applied to entire arrays at once instead of element-by-element loops
- Optimized C Implementation: NumPy functions are written in C, making them highly efficient.
- Less Overhead: Python lists store mixed data types, while NumPy arrays store homogeneous data, reducing memory and computation overhead
- Better Memory Usage: Data is stored in contiguous memory blocks, improving access speed

```
import numpy as np
import time
lst = list(range(1000000))
start = time.time()
lst = [x * 2 for x in lst]
print("Python Time:", time.time() - start)
arr = np.arange(1000000)
start = time.time()
arr = arr * 2
print("NumPy Time:", time.time() - start)
```

```
Python Time: 0.12811851501464844
NumPy Time: 0.005824089050292969
```

Q3.What types of mathematical operations can be performed using NumPy?

ANS:- NumPy supports a wide range of operations:

- Arithmetic operations: +, −, ×, ÷
- Statistical operations: mean, median, standard deviation.
- Linear algebra: dot product, matrix multiplication.
- Trigonometric functions: sin, cos, tan.
- Logarithmic & exponential functions

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print("Addition:", arr + 2)
print("Mean:", np.mean(arr))
print("Sine:", np.sin(arr))
print("Exponential:", np.exp(arr))
```

```
Addition: [3 4 5 6]
Mean: 2.5
Sine: [ 0.84147098 0.90929743 0.14112001 -0.7568025 ]
Exponential: [ 2.71828183 7.3890561 20.08553692 54.59815003]
```

Q4. Explain indexing and slicing in NumPy arrays.

ANS:-. Indexing and slicing allow accessing specific elements or subsets of arrays.

- Indexing: Access single elements
- Slicing: Extract portions of arrays

Example:-

- arr[0] → first element
- arr[1:4] → elements from index 1 to 3

NumPy also supports:

- Multi-dimensional indexing

- Boolean indexing

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50])
print("First Element:", arr[0])
print("Elements from index 1 to 3:", arr[1:4])
```

```
First Element: 10
Elements from index 1 to 3: [20 30 40]
```

Q5. How does NumPy integrate with libraries like:

ANS:- NumPy integrates seamlessly with other Python libraries, making it a core component of the data science ecosystem.

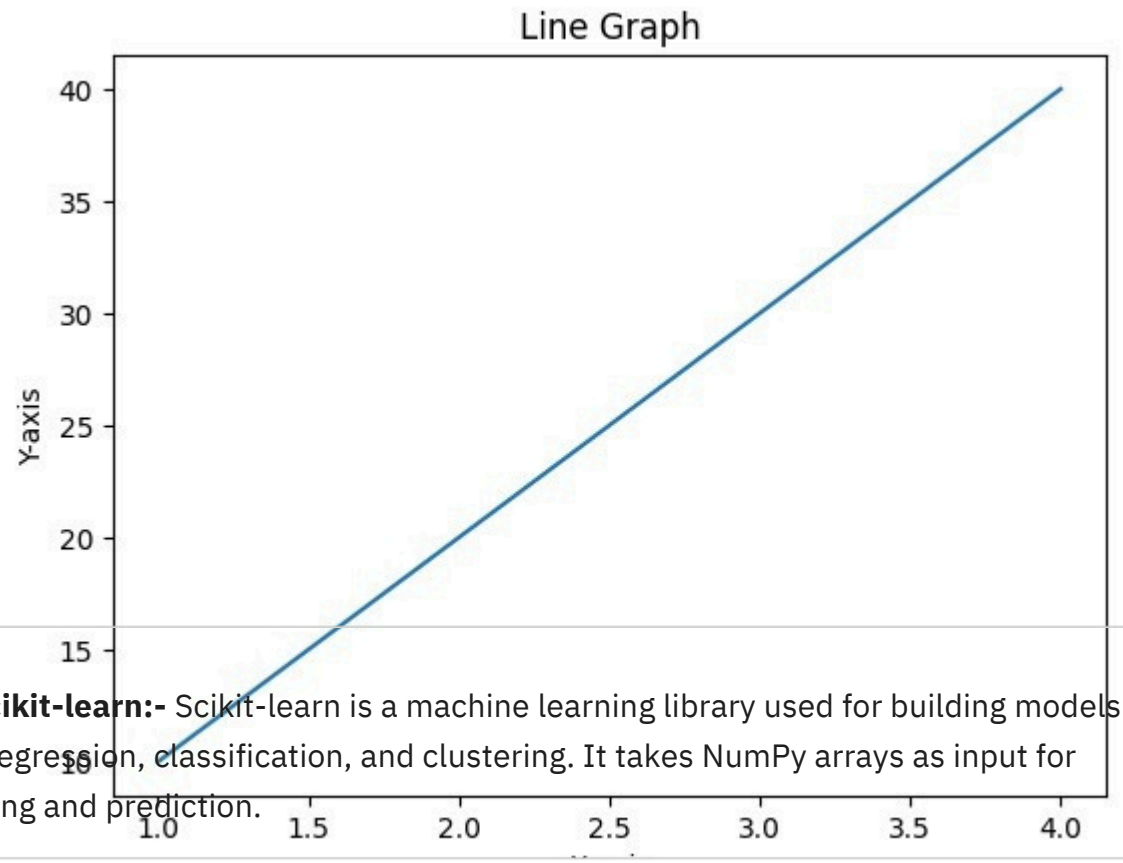
A) Pandas:- Pandas is a data analysis library that provides data structures like DataFrames and Series. It uses NumPy arrays internally for fast data processing and manipulation.

```
import numpy as np
import pandas as pd
data = np.array([10, 20, 30, 40])
df = pd.DataFrame(data, columns=["Values"])
print(df)
```

```
   Values
0      10
1      20
2      30
3      40
```

B) Matplotlib:- Matplotlib is a data visualization library used to create graphs, charts, and plots. It uses NumPy arrays to represent numerical data for plotting.

```
import numpy as np import
matplotlib.pyplot as plt x =
np.array([1, 2, 3, 4]) y =
np.array([10, 20, 30, 40])
plt.plot(x, y) plt.title("Line
Graph") plt.xlabel("X-axis")
plt.ylabel("Y-axis") plt.show()
```



C) Scikit-learn:- Scikit-learn is a machine learning library used for building models like regression, classification, and clustering. It takes NumPy arrays as input for training and prediction.

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = np.array([[1], [2], [3], [4]])
y = np.array([10, 20, 30, 40])
model = LinearRegression()
model.fit(X, y)
print("Prediction for 5:", model.predict([[5]]))
```

```
Prediction for 5: [50.]
```